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This document covers the deployment and integration of Sysmon on a Windows Server 2022 endpoint within the existing SOC lab. It documents the installation and configuration of Sysmon, collection of Sysmon event logs using the Splunk Universal Forwarder, and verification that the telemetry is successfully received by Splunk Enterprise.

The completed integration will provide enhanced endpoint visibility and form the foundation for future MITRE ATT&CK-based detection engineering and attack simulations.

Installation

First, I verified that Sysmon is not already installed, then I downloaded Sysmon from the official Microsoft website.

The screenshot is divided into two main horizontal sections. The top section shows a Windows PowerShell window titled 'Administrator: Windows PowerShell'. The text inside the window reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', 'Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows', 'PS C:\Windows\system32> Get-Service Sysmon*', and 'PS C:\Windows\system32>'. The bottom section shows a web browser window with the URL 'https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon'. The page is the 'Sysmon v15.22' download page. It features a left-hand navigation menu with a search bar and a list of items including 'PsLogList', 'RootkitRevealer', 'Sysmon' (which is highlighted), 'System Information', 'Miscellaneous', 'Sysinternals Suite', 'Microsoft Store', 'Community', 'Resources', 'Software License Terms', and 'Licensing FAQ'. The main content area of the page displays 'Sysmon v15.22' with a 'Summarize this article for me' button. Below this, it says 'By Mark Russinovich and Thomas Garnier' and 'Published: September 10, 2026'. There are two download links: 'Download Sysmon (2.8 MB)' and 'Download Sysmon for Linux (GitHub)'. On the right side, there is a section titled 'In this article' with a list of links: 'Introduction', 'Overview of Sysmon Capabilities', 'Screenshots', 'Usage', 'Examples', 'Events', 'Configuration files', 'Configuration Entries', and 'Event filtering entries'. The 'Introduction' link is currently selected.

Installation & Configuration

After successfully downloading Sysmon. Next I also downloaded the SwiftOnSecurity Sysmon Config, which is a community-developed Sysmon configuration that defines which system activities Sysmon should monitor and log. It provides a well-established baseline for collecting security-relevant events while reducing unnecessary noise. The configuration will be used as the starting point for the detection engineering lab instead of creating a custom Sysmon configuration from scratch.

The screenshot shows a web browser displaying the GitHub repository for `sysmon-config`. The repository is titled "sysmon-config | A Sysmon configuration file for everybody to fork". The README describes it as a Microsoft Sysinternals Sysmon configuration file template with default high-quality event tracing. It mentions that the file should function as a great starting point for system change monitoring in a self-contained and accessible package. The README also includes links to `sysmonconfig-export.xml`, `sysmon-modular` by @olafhartong, and `MalwareArchaeology Logging Cheat Sheets` by @HackerHurricane.

Below the browser window, a Windows File Explorer window is open, showing the contents of the `Sysmon` directory. The files listed are:

Name	Date modified	Type	Size
Eula	9/16/2026 11:07 AM	Text Document	8 KB
Sysmon	9/16/2026 11:07 AM	Application	6,018 KB
Sysmon64	9/16/2026 11:07 AM	Application	3,129 KB
Sysmon64a	9/16/2026 11:07 AM	Application	3,066 KB
sysmonconfig-export	9/16/2026 11:14 AM	XML Document	121 KB

Deployment

Sysmon v15.22 was successfully installed on the Windows Server 2022 endpoint using the SwiftOnSecurity configuration. The configuration was validated successfully, and both the Sysmon service and driver were started.

```

PS C:\Windows\system32> Get-Service Sysmon*
PS C:\Windows\system32> cd C:\Sysmon\Sysmon
PS C:\Sysmon\Sysmon> dir

Directory: C:\Sysmon\Sysmon

Mode                LastWriteTime         Length Name
----                -
-a----          9/16/2026 11:07 AM           7490 Eula.txt
-a----          9/16/2026 11:07 AM        6162240 Sysmon.exe
-a----          9/16/2026 11:07 AM        3203368 Sysmon64.exe
-a----          9/16/2026 11:07 AM        3138824 Sysmon64a.exe
-a----          9/16/2026 11:14 AM         123257 sysmonconfig-export.xml

PS C:\Sysmon\Sysmon> .\Sysmon64.exe -accepteula -i .\sysmonconfig-export.xml

System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

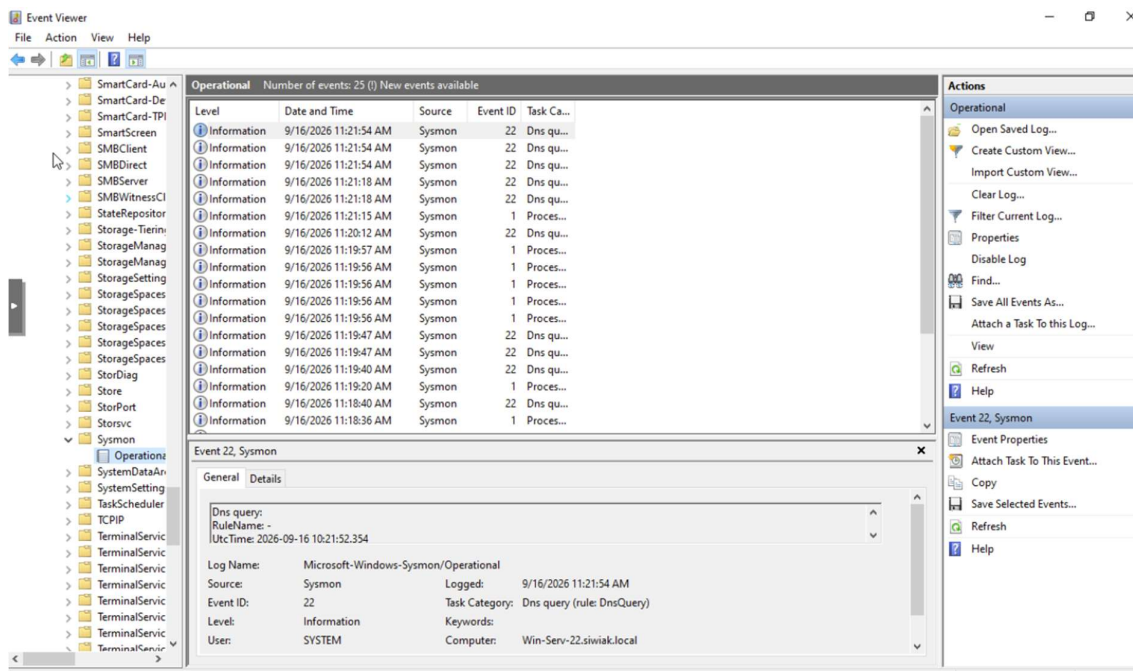
Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Sysmon\Sysmon> Get-Service Sysmon64

Status      Name      DisplayName
-----
Running     Sysmon64   Sysmon64

PS C:\Sysmon\Sysmon>

```

Sysmon event generation was verified locally using Windows Event Viewer. The Microsoft-Windows-Sysmon/Operational channel successfully recorded security telemetry, including Event ID 1 (Process Creation) and Event ID 22 (DNS Query), confirming that Sysmon was operational and the configuration was actively collecting events.



Sysmon → Splunk

The Splunk Universal Forwarder configuration was extended to monitor the Microsoft-Windows-Sysmon/Operational event channel. This allows Sysmon telemetry generated on the Windows Server endpoint to be collected and forwarded to the existing Splunk environment.

```
PS C:\> Get-Content "C:\Program Files\SplunkUniversalForwarder\etc\apps\SplunkUniversalForwarder\local\inputs.conf"

[WinEventLog::Application]
checkpointInterval = 5
current_only = 0
disabled = 0
start_from = oldest

[WinEventLog::Security]
checkpointInterval = 5
current_only = 0
disabled = 0
start_from = oldest

[WinEventLog::System]
checkpointInterval = 5
current_only = 0
disabled = 0
start_from = oldest

[perfmon://CPU Load]
counters = % Processor Time;% User Time
instances = _Total
interval = 10
object = Processor

[perfmon://Available Memory]
counters = Available Bytes
interval = 10
object = Memory

[perfmon://Free Disk Space]
counters = Free Megabytes;% Free Space
instances = _Total
interval = 3600
object = LogicalDisk

[perfmon://Network Interface]
counters = Bytes Received/sec;Bytes Sent/sec
instances = *
interval = 10
object = Network Interface
PS C:\> notepad "C:\Program Files\SplunkUniversalForwarder\etc\apps\SplunkUniversalForwarder\local\inputs.conf"
PS C:\>
```

*inputs.conf - Notepad

File Edit Format View Help

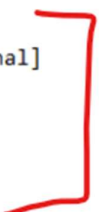
```
object = Processor

[perfmon://Available Memory]
counters = Available Bytes
interval = 10
object = Memory

[perfmon://Free Disk Space]
counters = Free Megabytes;% Free Space
instances = _Total
interval = 3600
object = LogicalDisk

[perfmon://Network Interface]
counters = Bytes Received/sec;Bytes Sent/sec
instances = *
interval = 10
object = Network Interface

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
checkpointInterval = 5
current_only = 0
disabled = 0
start_from = oldest
```

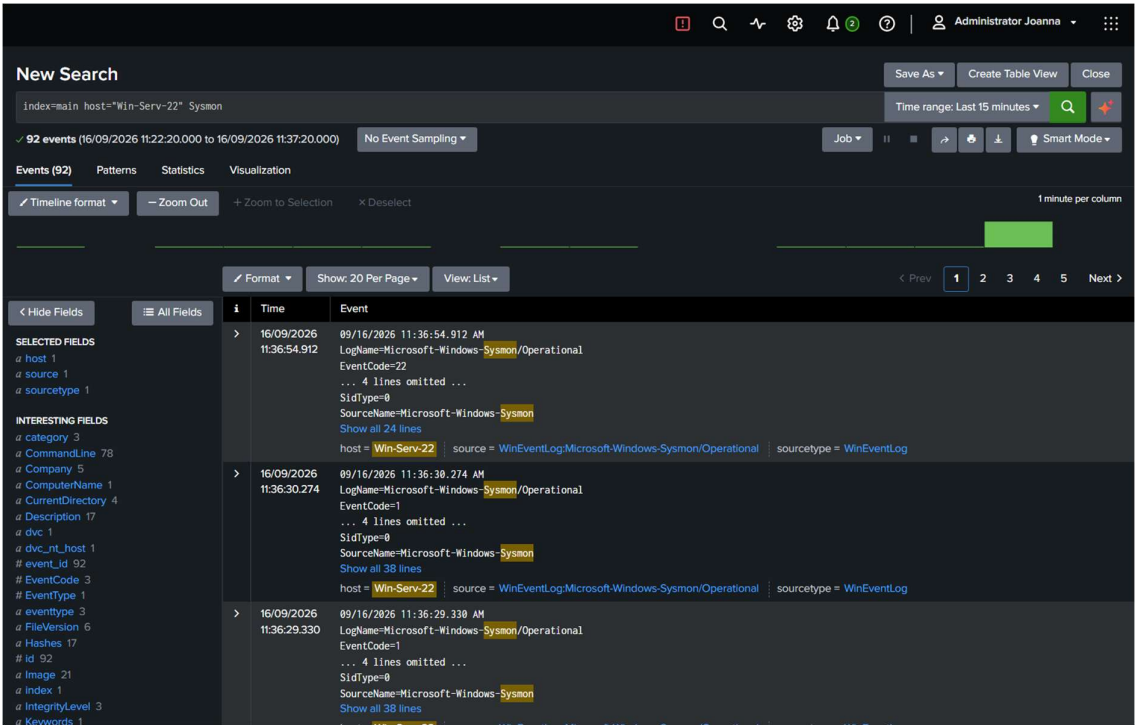


```
PS C:\> Restart-Service SplunkForwarder
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
PS C:\> Get-Service SplunkForwarder
```

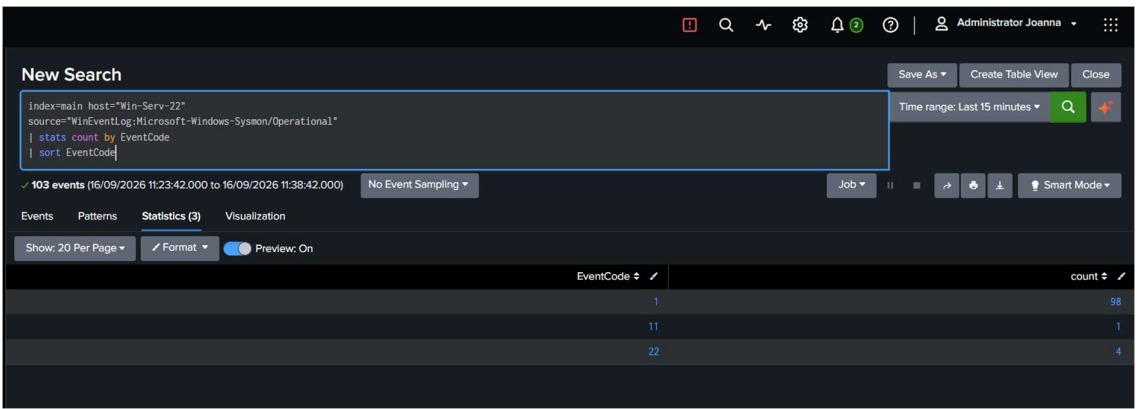
Status	Name	DisplayName
Running	SplunkForwarder	SplunkForwarder

Summary

Sysmon telemetry was successfully received by Splunk Enterprise from Win-Serv-22. The results confirm end-to-end collection through the Splunk Universal Forwarder, including Sysmon Event ID 1 (Process Creation) and Event ID 22 (DNS Query).



A final SPL verification was performed to identify the Sysmon event types being ingested by Splunk. During the observed period, Splunk received Process Creation (Event ID 1), File Creation (Event ID 11), and DNS Query (Event ID 22) telemetry from Win-Serv-22. This confirmed successful end-to-end Sysmon log collection and provided the telemetry required for subsequent detection engineering activities.



This integration establishes the endpoint telemetry foundation for the next phase of the project, where Sysmon data will be used to develop, test and validate MITRE ATT&CK-mapped detections in Splunk